

Network Function Virtualization

Programmable Networks A.Y. 2024/25

Outline

- Introduction
- NFV Architecture
- Virtualization
- Resource Allocation
- Creation of a Network Service

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Introduction

- In an enterprise network, middleboxes are ubiquitous and their number is comparable with the number of routers and switches needed to maintain the operation of the network

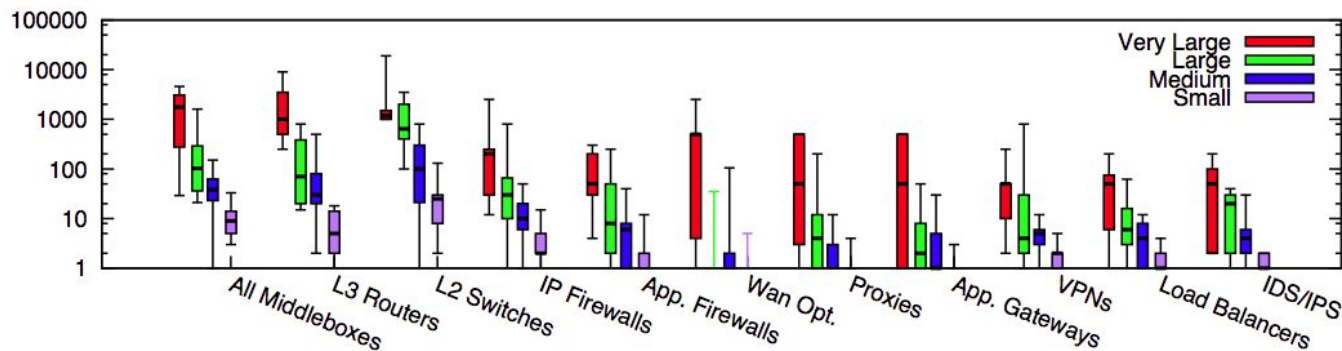
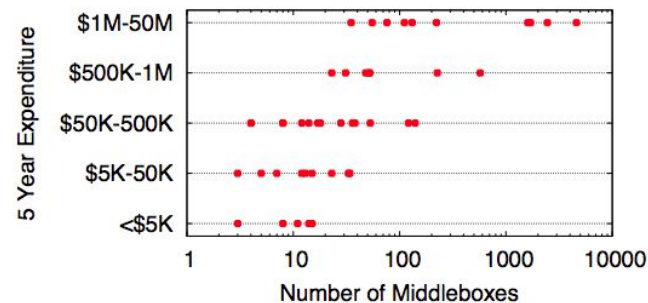


Figure 1: Box plot of middlebox deployments for small (fewer than 1k hosts), medium (1k-10k hosts), large (10k-100k hosts), and very large (more than 100k hosts) enterprise networks. Y-axis is in log scale.

“Making Middleboxes Someone Else’s Problem: Network Processing as a Cloud Service”

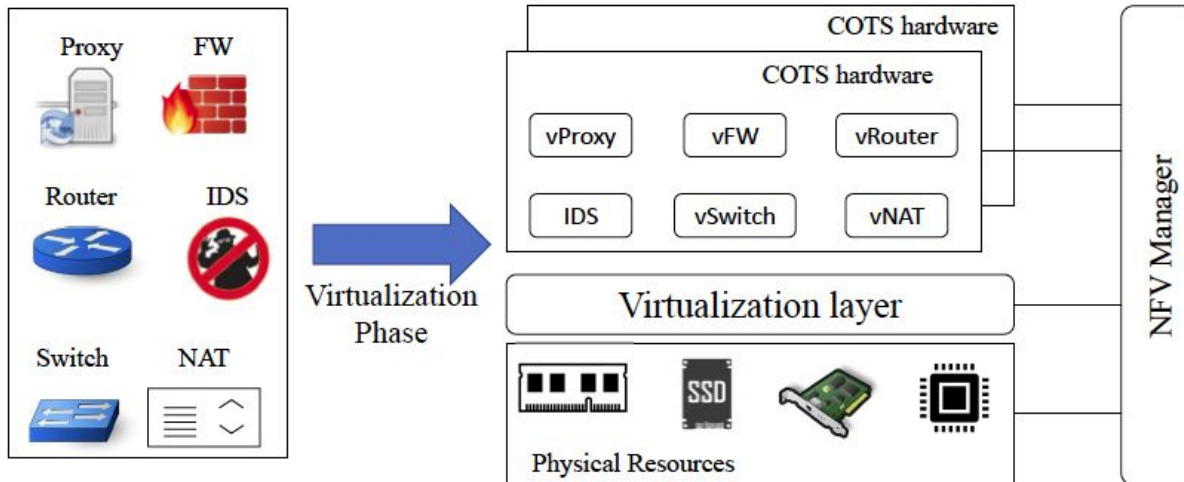
Introduction

- Middleboxes have several **shortcomings**
 - expensive
 - require specialized managing personnel
 - high delay in introducing new networking features
 - high energy cost
 - short life cycles
 - have vendor specific interfaces
 - slow protocol standardization

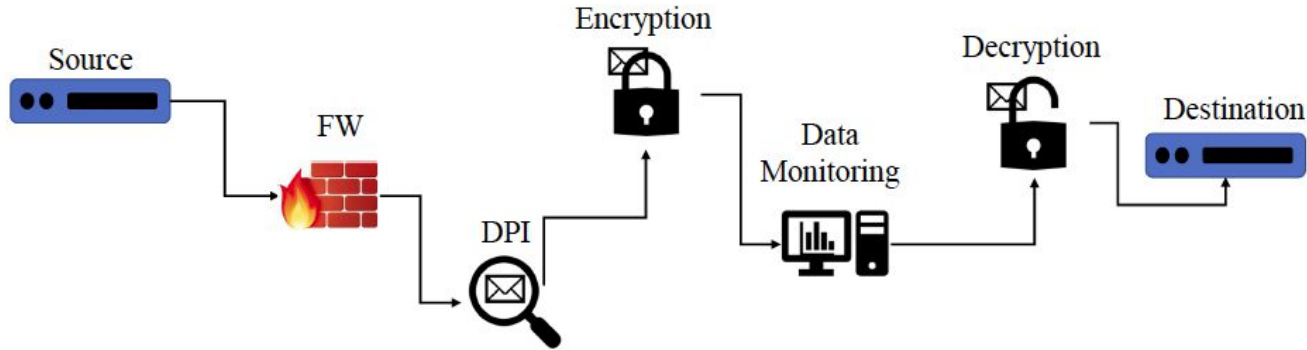


The NFV Paradigm

- **NFV** is the term used to represent the implementation of hardware appliances providing data plane **Network Functions (NFs) in software**
- NFs are **executed on Commodity Off-The-Shelf** (COTS) hosts (e.g., x86-based blades), by leveraging on virtualization technology



Service Function Chains



- **Virtual functions are building blocks** that can be interconnected to implement complex NSs
- A NS requires data flows to pass through a certain fixed set of middleboxes in a specific order
- The task of **steering the traffic among the middleboxes** needed to realize service chains is commonly known as middlebox orchestration

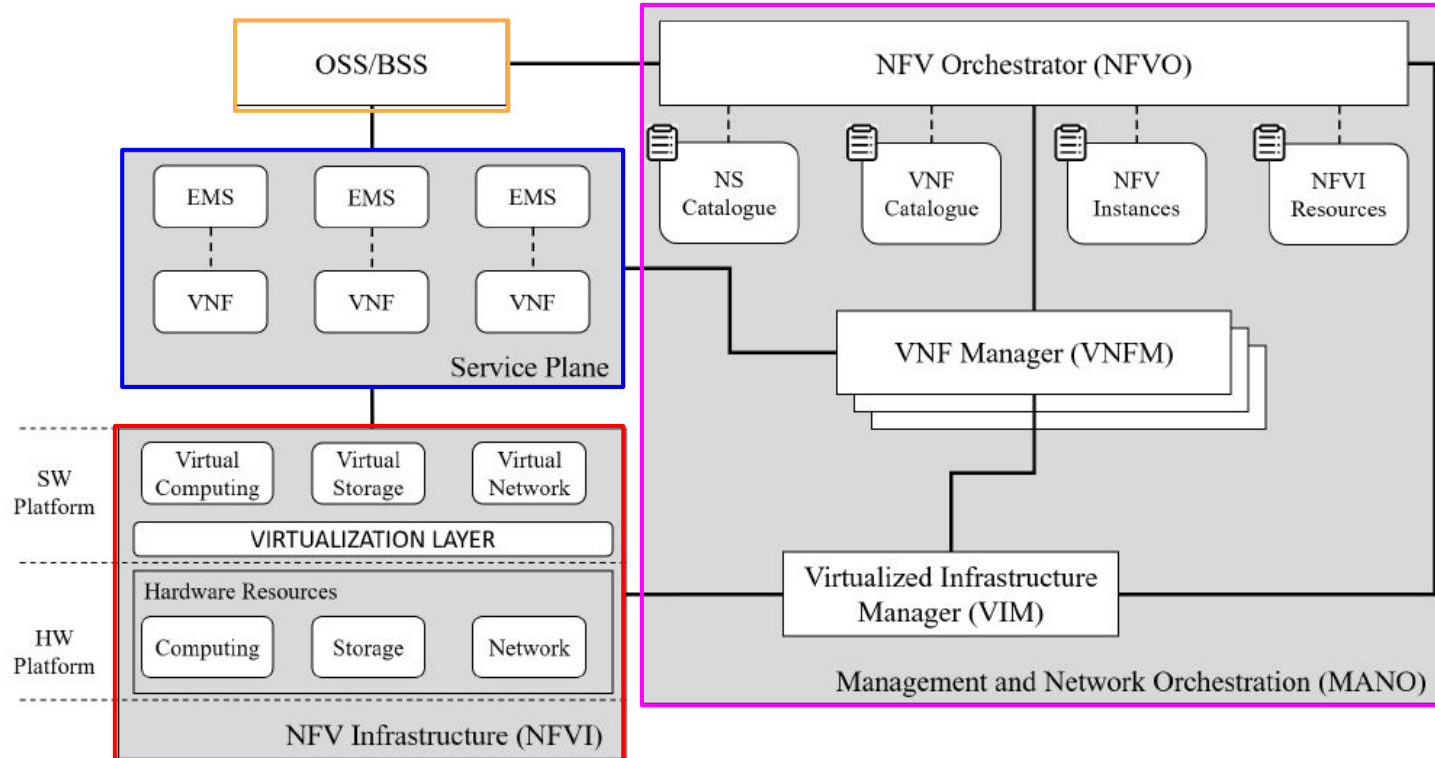
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NFV Architecture

- The main architectural components are
 - Network Function Virtualization Infrastructure (NFVI)
 - Service Plane
 - Management and Network Orchestration (MANO)
 - Operations Support System (OSS)
 - service provisioning, network inventory, network configuration, and fault management
 - Business Support System (BSS)
 - ordering, billing, and revenue management aspects

NFV Architecture



NFV Infrastructure (NFVI)

- The **NFV Infrastructure (NFVI) is a network of nodes**, each virtualizing **computation, storage, and networking**, and able to host VNFs
 - each NFVI node is called **NFVI Point-of Presence (NFVI-PoP)**
- NFVI-PoP are interconnected through a transport network
- The NFVI is composed of two main elements
 - The **hardware platform** is made of all the resources related to the network infrastructure, as well as computing and storage resources
 - The **software platform** is constituted by the
 - Virtualized resources - abstractions of hardware components
 - Virtualization layer - decouple software from the hardware

Service Plane

- The **Service Plane is populated by VNFs** that can be provided alone, or chained to realize more complex Network Services (NS)
- A VNF or a whole service chain can be
 - mapped to a single VM
 - deployed across multiple NFVI-PoPs
- The northbound interface of a VNF is represented by an **Element Management System (EMS)**
 - provides VNF users with the same interface as if it were a physical NF

NFV MAnagement and Network Orchestration (MANO)

- The MANO component allows for **fast, scalable and dynamic composition and allocation of VNFs to execute a NS**
- The main objectives of NFV MANO are
 - how to compose VNFs for a given NS
 - how to efficiently allocate and schedule the VNFs of a NS inside the NFVI PoPs of the underlying NFVI
- MANO also provides centralized control over service provisioning and management

NFV MANO

- MANO consists of three main parts:
 - **Virtualized Infrastructure Manager (VIM):**
 - it takes care of the **control of NFVI computing, storage, and networking** resources
 - A de-facto standard for VIM is OpenStack
 - **VNF Managers (VNFM):**
 - VNFMs **manage the lifecycle of the associated VNF instances** by initializing, updating, querying, scaling and terminating them
 - Each VNF instance has an associated VNFM
 - **NFV Orchestrator (NFVO)**
 - it achieves the **orchestration of NFVI resources across multiple VIMs** and instantiates all the VNFMs
 - manages the lifecycle of network services

NFV MANO

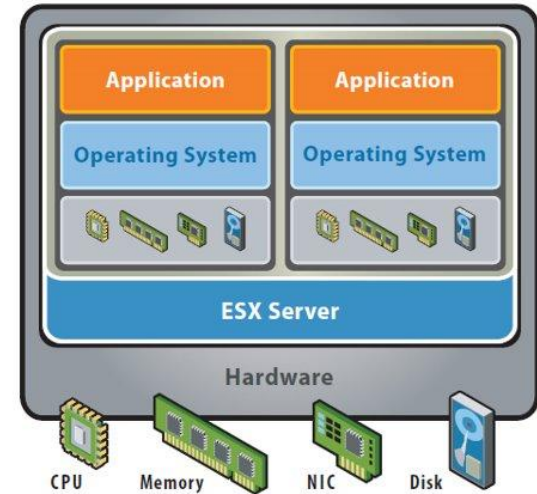
- Four catalogues are included in the MANO
 - NS Catalogue
 - represents the repository of all the on-boarded NSs
 - VNF Catalogue
 - represents the repository of all the on-boarded VNF Packages, which are tar-archives containing all the necessary information to manage the lifecycle of the VNFs
 - NFV Instances repository
 - holds information about all VNF and NS instances during their lifecycles
 - NFVI Resources repository
 - holds information about available/reserved/allocated NFVI resources

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What and Why

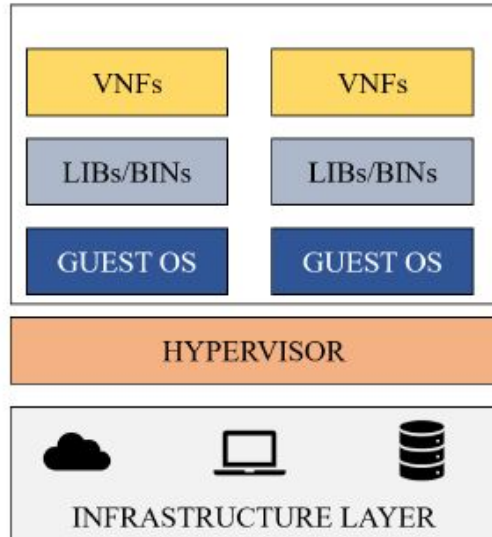
- Virtualization is a **technology used to create virtual representations of physical resources** such as CPU, storage and networking
- Main advantages are
 - **Sharing** - break up a large resource
 - **Isolation** - protection from other tenants
 - **Aggregation** - combine many resources into one
 - **Dynamics** - fast allocation
 - **Ease of Management** - easy distribution, deployment, testing



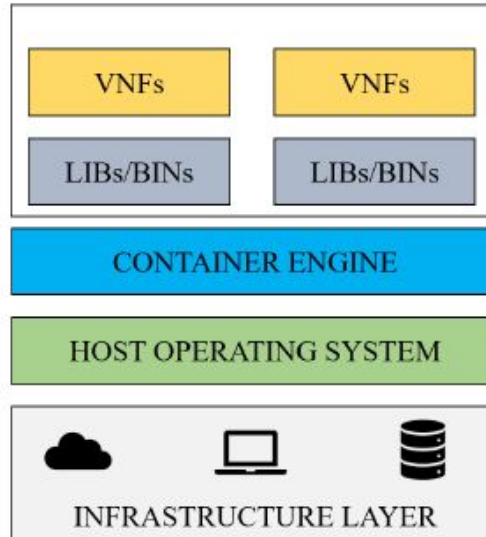
VMware ESX Server virtualizes server storage and networking, allowing multiple applications to run in virtual machines on the same physical server.

Virtualization of NFs

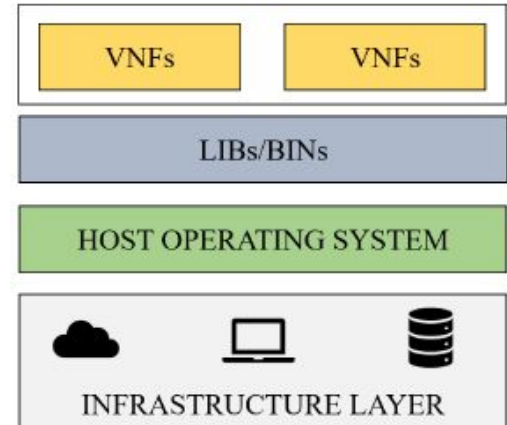
VIRTUAL MACHINE DEPLOYMENT



CONTAINER DEPLOYMENT



BARE-METAL PROCESS DEPLOYMENT

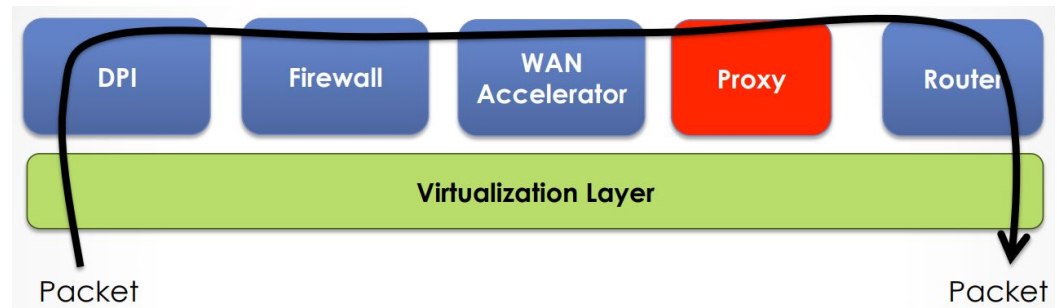
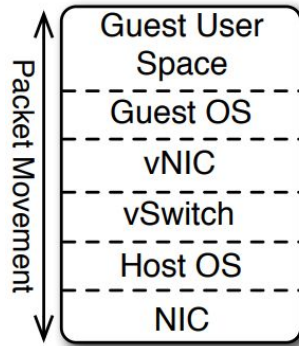


VNF deployment inside Virtual Machines (VMs)

- Most adopted solution to run VNFs leverages **KVM-based VMs**
 - Kernel-based Virtual Machine (KVM) is a virtualization technology built into Linux
 - KVM lets you turn Linux into a hypervisor
- The **hypervisor is a middleware that offers a virtual platform to accommodate a variety of guest operating systems**
- Hypervisors are classified as:
 - **native hypervisors**
 - installed on the host machine hardware directly (Xen, VMware ESX)
 - **hosted hypervisors**
 - executed inside a traditional operating system installed on the host machine (KVM, Virtualbox, Parallel Desktop, etc.)

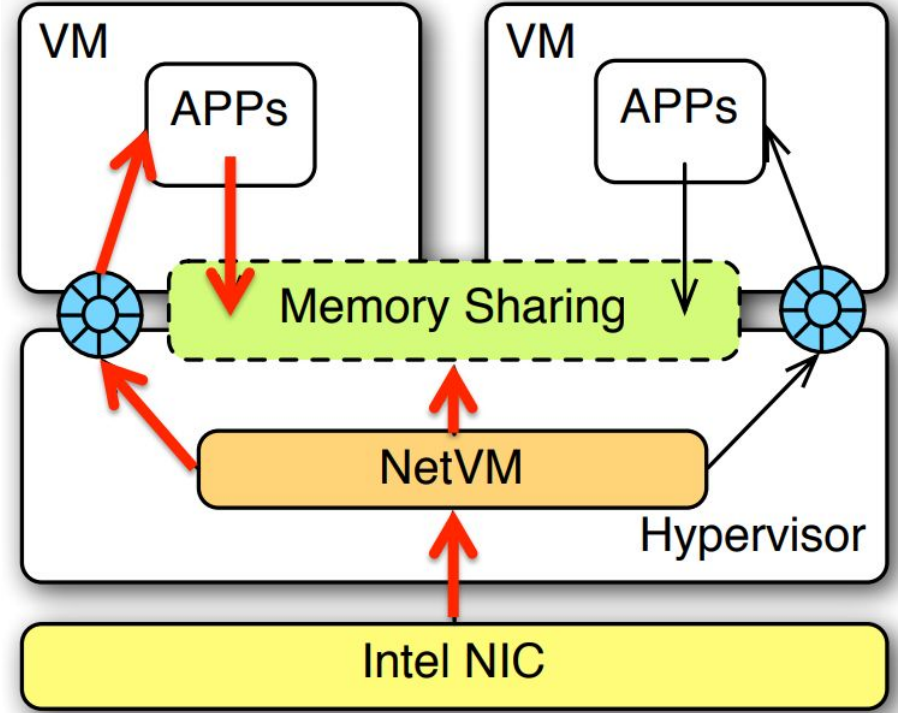
NetVM

- **NetVM** is one of the most popular **VM-based NFV platforms**
- It is an **NFV environment built over the KVM platform and the Intel DPDK library**
- It avoids the overhead introduced by virtualization when moving packets between VMs instantiated in the same host machine



NetVM

- NetVM (with DPDK) runs in hypervisor User Space
- For zero packet copy, memory is shared for network data
- Each VM has its own ring to receive/transmit a packet descriptor



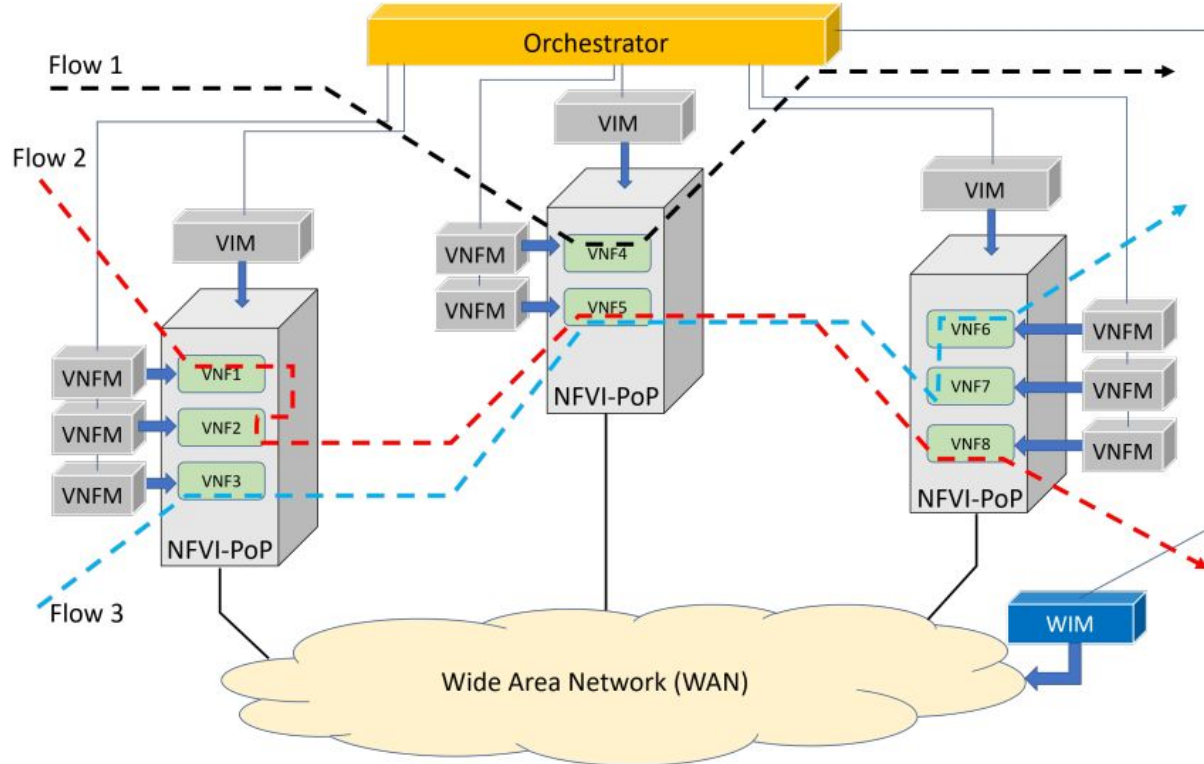
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Resource Allocation

- Resource allocation in NFV consists in determining
 - the **optimal placement of VNFs**
 - the **optimal migration strategies** to migrate functions from one server to another
- Optimization is accomplished with **different objectives** in mind
 - e.g., load balancing, reduction of CAPEX and OPEX, energy saving, recovery from failures, etc.
- NFVO is responsible for resource allocation

Resource Allocation: an example

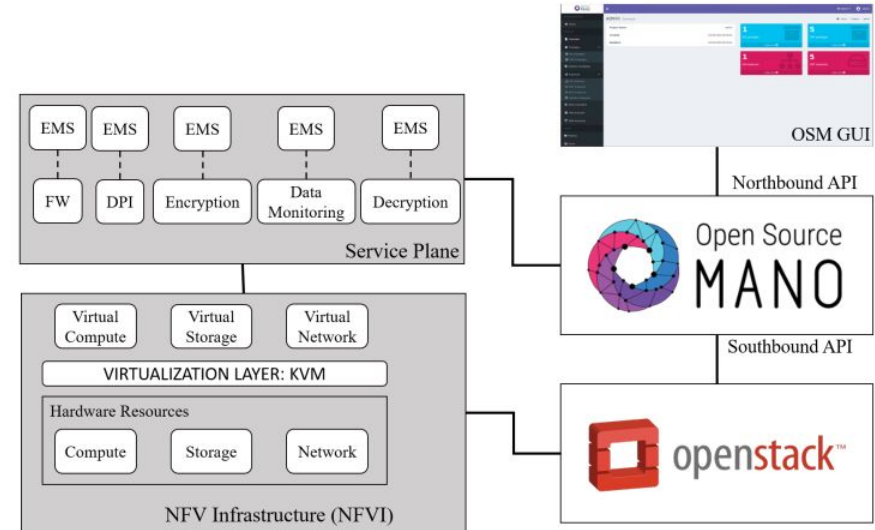


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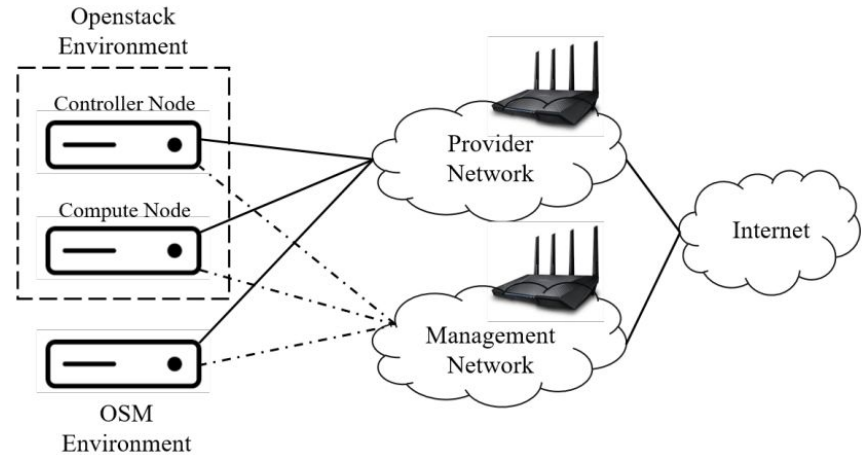
Creation of a Network Service

- The figure shows a deployment of the NFV Architecture, in which
 - OSM is used as NFVO and VNFM
 - OpenStack is used as VIM
- OSM has a Northbound interface with the APIs that allow the use of an OSM GUI
- in the Southbound interface there are the APIs needed to interact with OpenStack
- The targeted Network Service is composed of the following functions:
 - Firewall, DPI, Encryption, Data Monitoring, and Decryption



Creation of a Network Service

- The OpenStack Environment is composed by two nodes:
 - **Controller** - manages all the services that are needed in the OpenStack Environment
 - **Compute** - runs the hypervisor to run instances and the Networking service agent to provide connectivity to them
- The OSM Environment is constituted by a single node in which the OSM platform is deployed



Creation of a Network Service

- To add an OpenStack VIM through the OSM CLI:

```
osm vim-create --name openstack-site --user admin --password userpwd --auth url http://10.10.10.11:5000/v2.0
--tenant admin --account type openstack
```

- To create the NS, the first step consists in the upload of the VNF images inside OpenStack (through the OpenStack CLI)

```
openstack image create "name of image" --file name of the file --disk-format qcow2 --container-format base
--public
```

Images

Q Click here for filters. X							
Displaying 18 items							
☐	Owner	Name	Type	Status	Visibility	Protected	Disk Format
☐	admin	Decryption	Image	Active	Public	No	QCOW2
☐	admin	DataMonitoring	Image	Active	Public	No	QCOW2
☐	admin	Encryption	Image	Active	Public	No	QCOW2
☐	admin	DPI	Image	Active	Public	No	QCOW2
☐	admin	Firewall	Image	Active	Public	No	QCOW2

Creation of a Network Service

- The second step regards the creation of the **VNF Descriptors** (VNFD) of each VNF composing the NS, and the **NS Descriptor** (NSD) of the NS
 - The descriptors are **configuration models written in YAML or JSON**
 - The VNFD, it contains information about a single VNF
 - It is used by the VNFM during the instantiation process and the management lifecycle of the VNF
 - The NSD is used to describe in detail the structure of the NS
- Once the descriptors are created, it is possible to upload them inside OSM

```
osm vnfd-create name of vnfd.tar.gz
```

```
osm nsd-create name of nsd.tar.gz
```

Creation of a Network Service

Edit VNFD Descriptor

YAML

JSON

```
1  vnfd:vnfd-catalog:
2    vnfd:
3      - connection-point:
4        - name: eth0
5          type: VPORT
6        - name: eth1
7          type: VPORT
8      description: Firewall VNF
9      id: firewall_vnfd
10     logo: fw.png
11     mgmt-interface:
12       cp: eth0
13     name: firewall_vnf
14     short-name: firewall_vnf
15     vdu:
16       - description: firewall_vnfd-VM
17         id: firewall_vnfd-VM
18         image: Firewall
```

```
19
20     interface:
21       - external-connection-point-ref: eth0
22         name: eth0
23         type: EXTERNAL
24         virtual-interface:
25           bandwidth: '0'
26           type: VIRTIO
27       - external-connection-point-ref: eth1
28         name: eth1
29         type: EXTERNAL
30         virtual-interface:
31           bandwidth: '0'
32           type: VIRTIO
33     name: firewall_vnfd-VM
34     vm-flavor:
35       memory-mb: 1024
36       storage-gb: 3
37       vcpu-count: 2
38     vendor: OSM
39     version: '1.0'
```

Creation of a Network Service

Edit NSD Descriptor

YAML

JSON

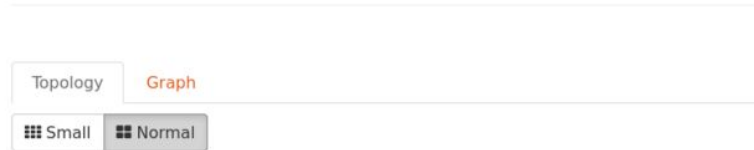
```
1 nsd:nsd-catalog:
2   nsd:
3     - constituent-vnfd:
4       - member-vnf-index: 1
5         vnfd-id-ref: firewall_vnfd
6       - member-vnf-index: 2
7         vnfd-id-ref: dpi_vnfd
8       - member-vnf-index: 3
9         vnfd-id-ref: encryption_vnfd
10      - member-vnf-index: 4
11        vnfd-id-ref: dataMonitoring_vnfd
12      - member-vnf-index: 5
13        vnfd-id-ref: decryption_vnfd
14      description: ns_security descriptor
15      id: ns_security
16      name: ns_security
17      short-name: ns_security
18      vendor: OSM Composer
19      version: '1.0'
20      vld:
21        - id: provider
22          mgmt-network: 'true'
23          name: provider
24          type: ELAN
25          vnfd-connection-point-ref:
26            - member-vnf-index-ref: '1'
27              vnfd-connection-point-ref: eth0
28              vnfd-id-ref: firewall_vnfd
29            - member-vnf-index-ref: '2'
30              vnfd-connection-point-ref: eth0
31              vnfd-id-ref: dpi_vnfd
32            - member-vnf-index-ref: '3'
33              vnfd-connection-point-ref: eth0
34              vnfd-id-ref: encryption_vnfd
```

```
35     - member-vnf-index-ref: '4'
36       vnfd-connection-point-ref: eth0
37       vnfd-id-ref: dataMonitoring_vnfd
38     - member-vnf-index-ref: '5'
39       vnfd-connection-point-ref: eth0
40       vnfd-id-ref: decryption_vnfd
41   - id: ns_security_1
42     name: ns_security_1
43     type: ELAN
44     vnfd-connection-point-ref:
45       - member-vnf-index-ref: '1'
46         vnfd-connection-point-ref: eth1
47         vnfd-id-ref: firewall_vnfd
48       - member-vnf-index-ref: '2'
49         vnfd-connection-point-ref: eth1
50         vnfd-id-ref: dpi_vnfd
51   - id: ns_security_2
52     name: ns_security_2
53     type: ELAN
54     vnfd-connection-point-ref:
55       - member-vnf-index-ref: '2'
56         vnfd-connection-point-ref: eth2
57         vnfd-id-ref: dpi_vnfd
58       - member-vnf-index-ref: '3'
```

```
59       vnfd-connection-point-ref: eth1
60       vnfd-id-ref: encryption_vnfd
61   - id: ns_security_3
62     name: ns_security_3
63     type: ELAN
64     vnfd-connection-point-ref:
65       - member-vnf-index-ref: '3'
66         vnfd-connection-point-ref: eth2
67         vnfd-id-ref: encryption_vnfd
68       - member-vnf-index-ref: '4'
69         vnfd-connection-point-ref: eth1
70         vnfd-id-ref: dataMonitoring_vnfd
71   - id: ns_security_4
72     name: ns_security_4
73     type: ELAN
74     vnfd-connection-point-ref:
75       - member-vnf-index-ref: '4'
76         vnfd-connection-point-ref: eth2
77         vnfd-id-ref: dataMonitoring_vnfd
78       - member-vnf-index-ref: '5'
79         vnfd-connection-point-ref: eth1
80         vnfd-id-ref: decryption_vnfd
81
```

Creation of a Network Service

Network Topology



- 1 Firewall
- 2 DPI
- 3 Encryption
- 4 Data Monitoring
- 5 Decryption

